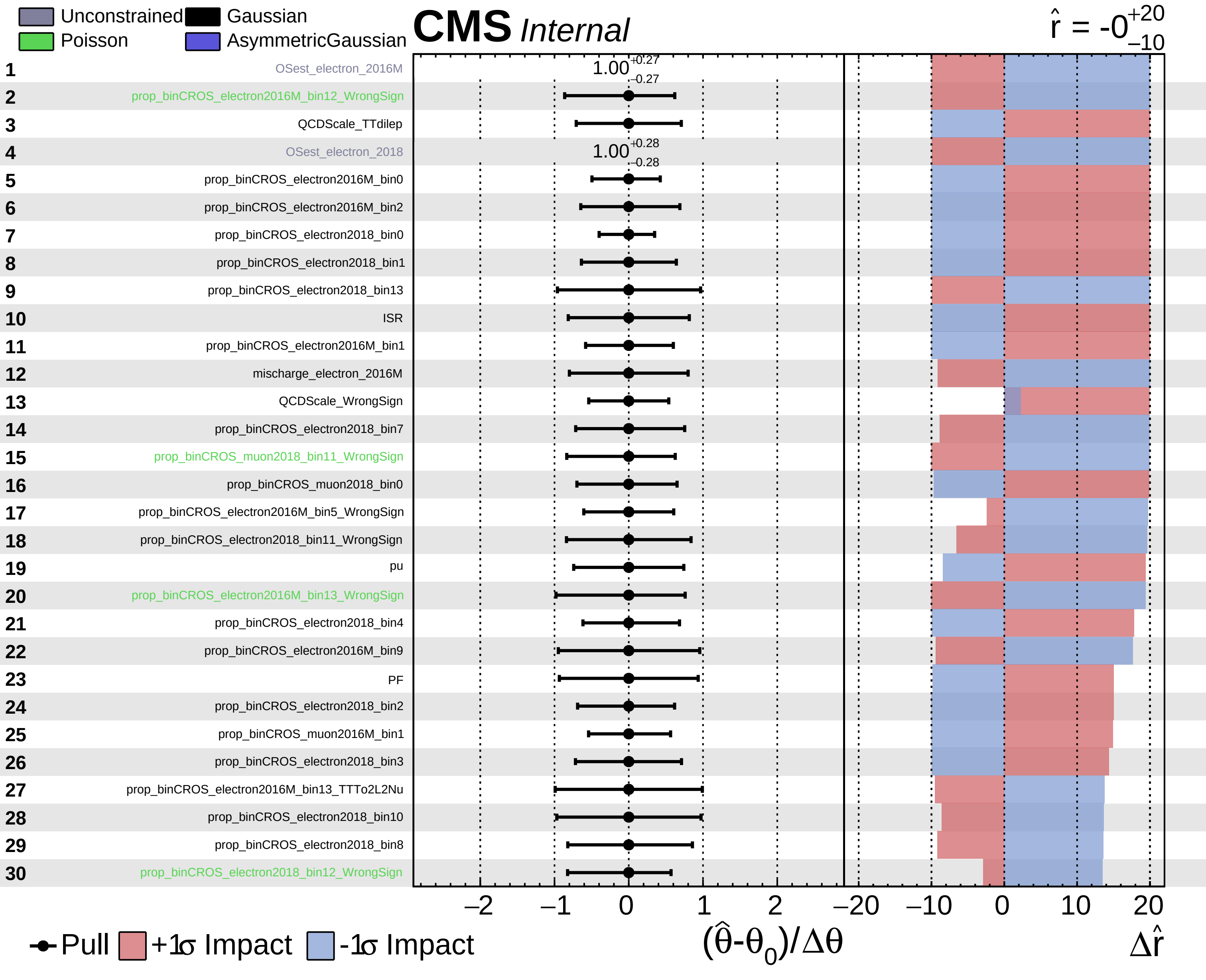
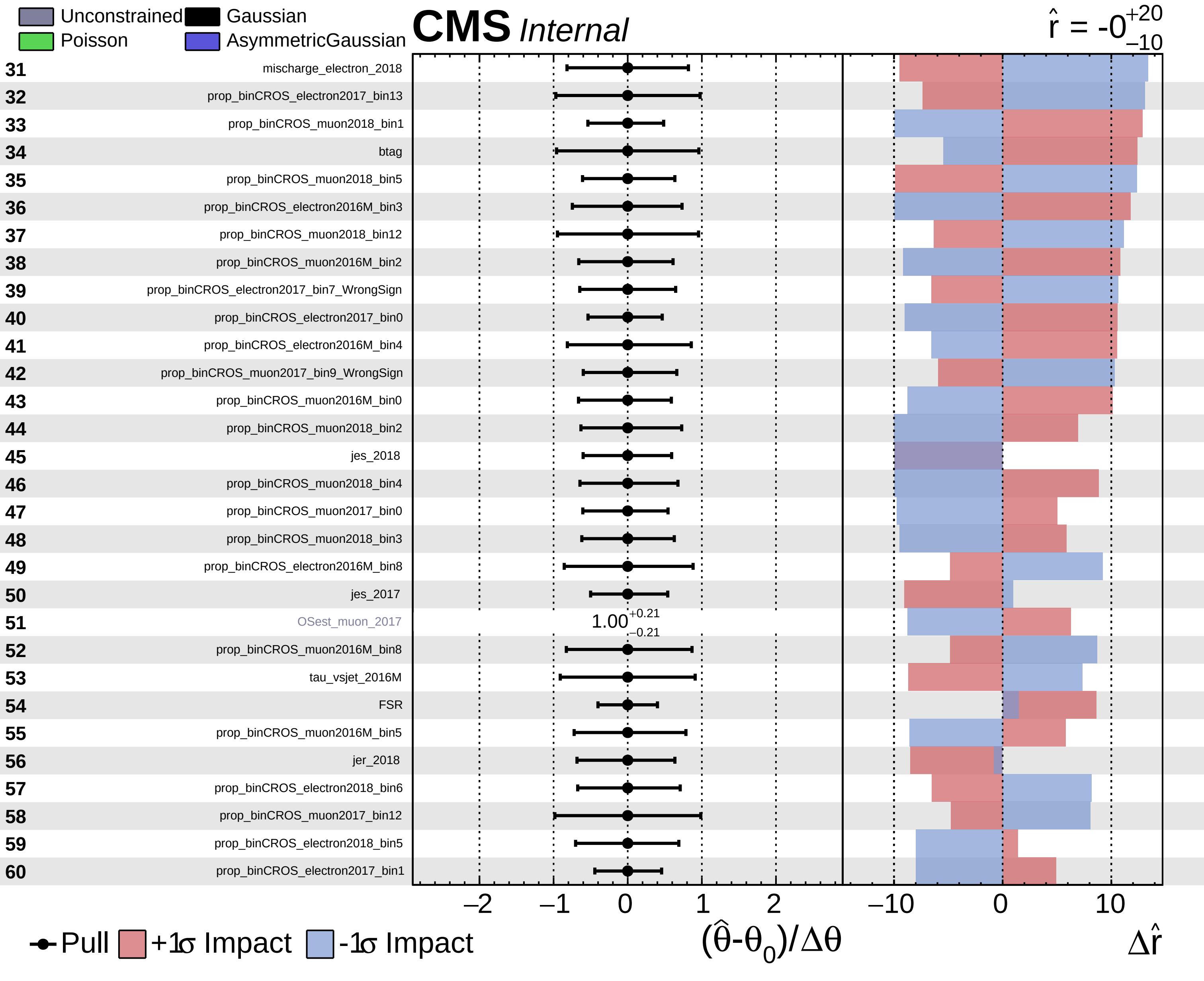
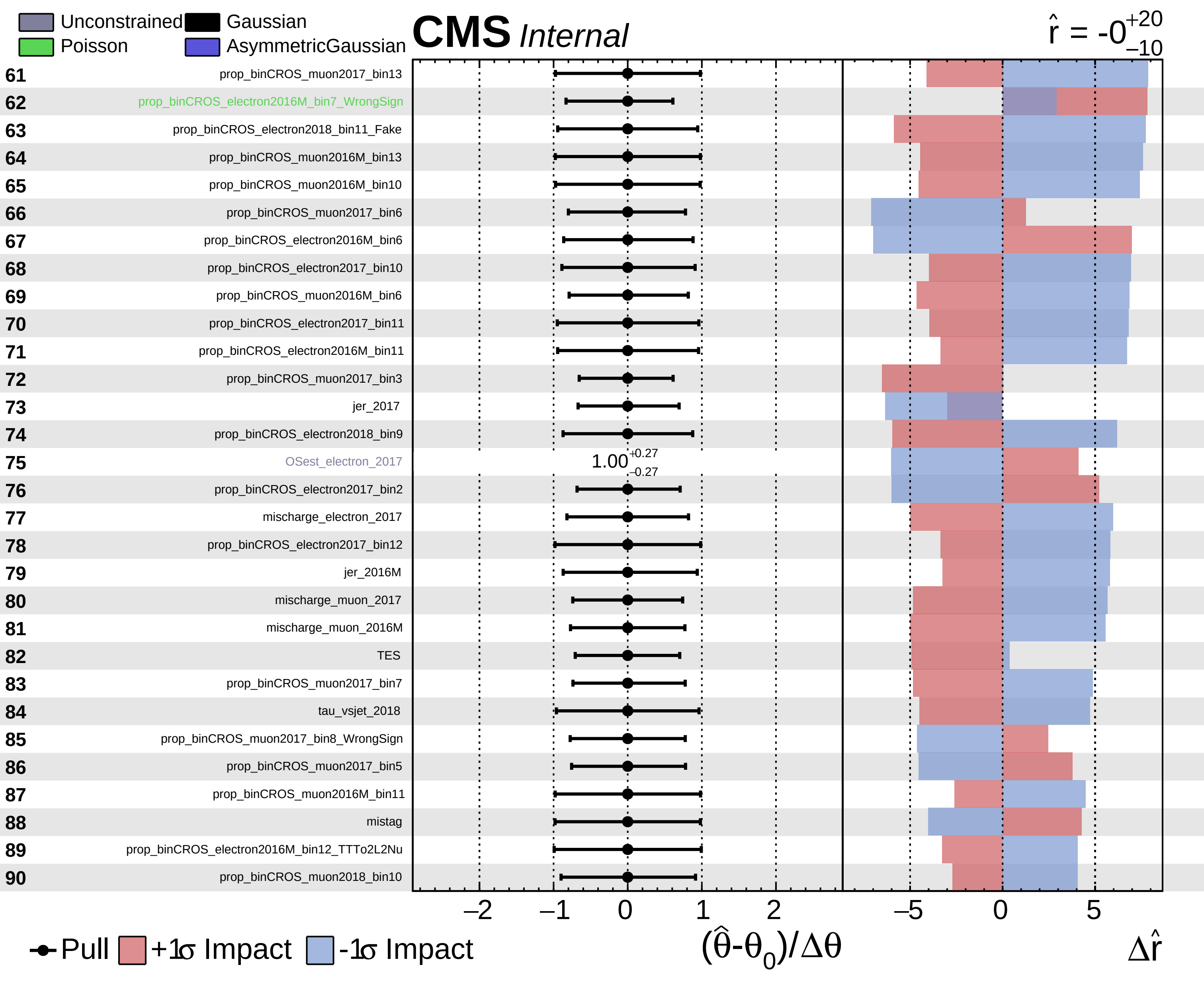
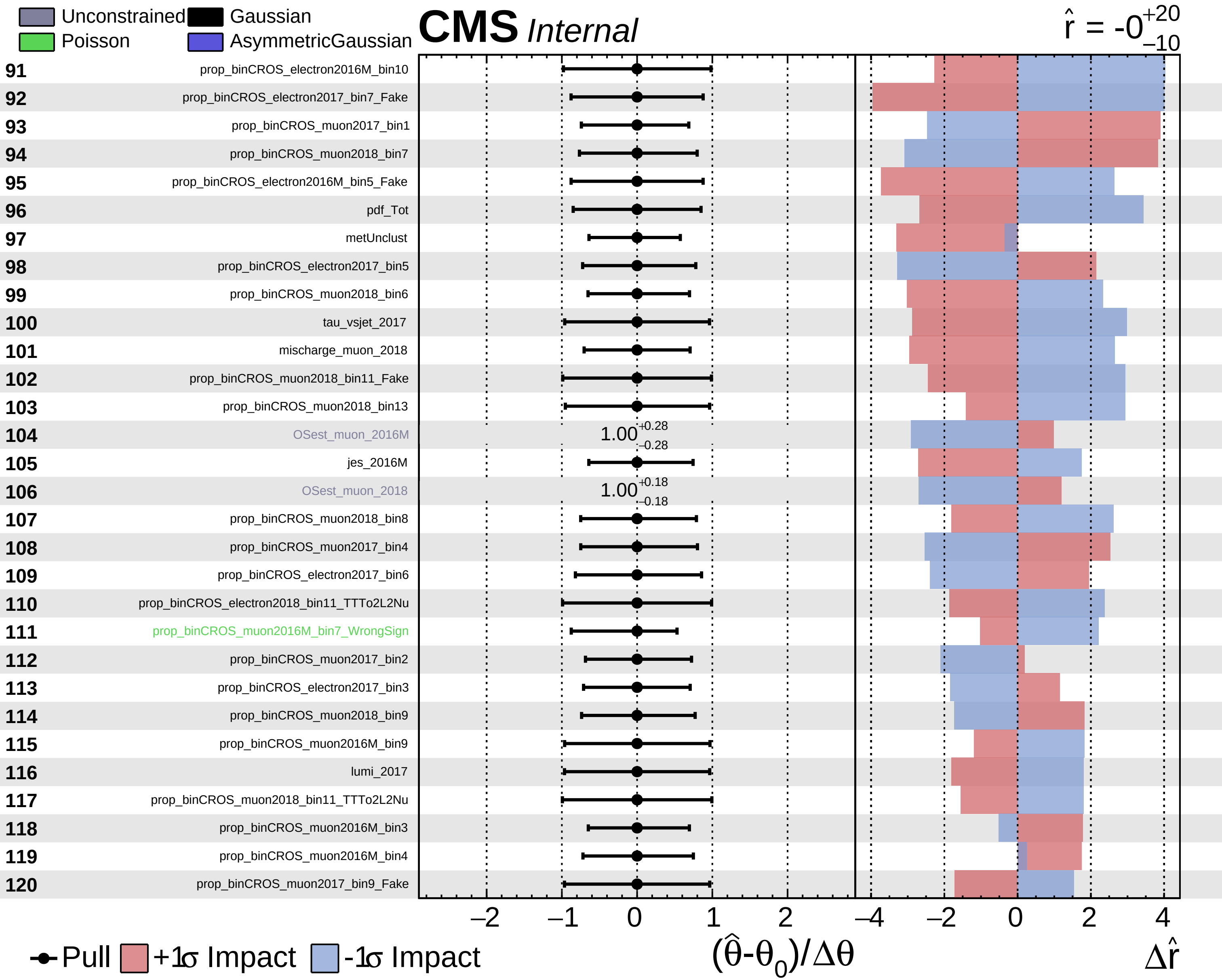


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0 +20 −10
1 OSest_electron_2016M 1.00 +0.27 −0.27
2 prop_binCROS_electron2016M_bin12_WrongSign
3 QCDScale_TTdilep
4 OSest_electron_2018 1.00 +0.28 −0.28
5 prop_binCROS_electron2016M_bin0
6 prop_binCROS_electron2016M_bin2
7 prop_binCROS_electron2018_bin0
8 prop_binCROS_electron2018_bin1
9 prop_binCROS_electron2018_bin13
10 ISR
11 prop_binCROS_electron2016M_bin1
12 mischarge_electron_2016M
13 QCDScale_WrongSign
14 prop_binCROS_electron2018_bin7
15 prop_binCROS_muon2018_bin11_WrongSign
16 prop_binCROS_muon2018_bin0
17 prop_binCROS_electron2016M_bin5_WrongSign
18 prop_binCROS_electron2018_bin11_WrongSign
19 pu
20 prop_binCROS_electron2016M_bin13_WrongSign
21 prop_binCROS_electron2018_bin4
22 prop_binCROS_electron2016M_bin9
23 PF
24 prop_binCROS_electron2018_bin2
25 prop_binCROS_muon2016M_bin1
26 prop_binCROS_electron2018_bin3
27 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
28 prop_binCROS_electron2018_bin10
29 prop_binCROS_electron2018_bin8
30 prop_binCROS_electron2018_bin12_WrongSign
−2 −1 0 1 2
−20 −10 0 10 20
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

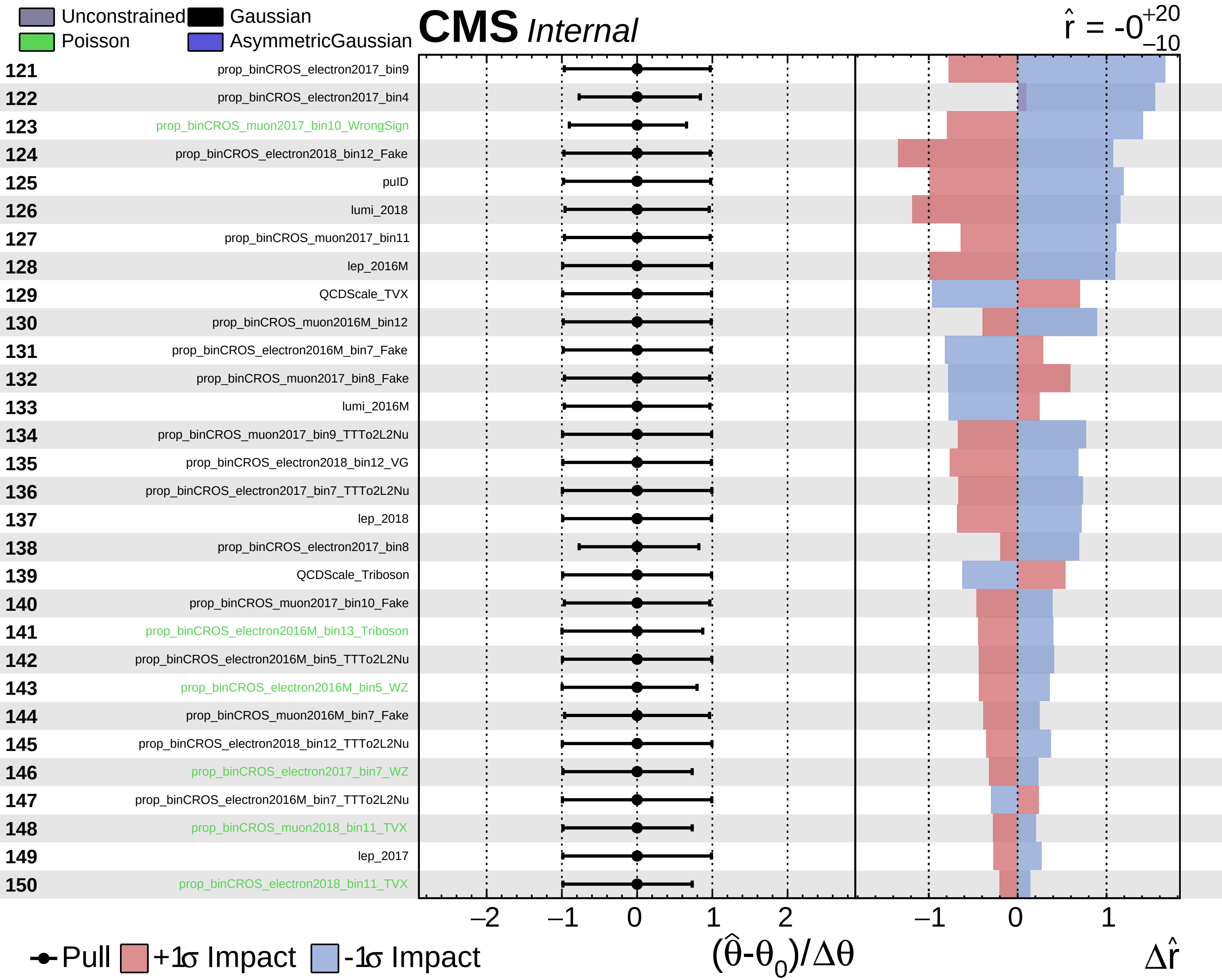

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
31 mischarge_electron_2018
32 prop_binCROS_electron2017_bin13
33 prop_binCROS_muon2018_bin1
34 btag
35 prop_binCROS_muon2018_bin5
36 prop_binCROS_electron2016M_bin3
37 prop_binCROS_muon2018_bin12
38 prop_binCROS_muon2016M_bin2
39 prop_binCROS_electron2017_bin7_WrongSign
40 prop_binCROS_electron2017_bin0
41 prop_binCROS_electron2016M_bin4
42 prop_binCROS_muon2017_bin9_WrongSign
43 prop_binCROS_muon2016M_bin0
44 prop_binCROS_muon2018_bin2
45 jes_2018
46 prop_binCROS_muon2018_bin4
47 prop_binCROS_muon2017_bin0
48 prop_binCROS_muon2018_bin3
49 prop_binCROS_electron2016M_bin8
50 jes_2017
51 OSest_muon_2017
$1.00^{+0.21}_{-0.21}$
52 prop_binCROS_muon2016M_bin8
53 tau_vsjet_2016M
54 FSR
55 prop_binCROS_muon2016M_bin5
56 jer_2018
57 prop_binCROS_electron2018_bin6
58 prop_binCROS_muon2017_bin12
59 prop_binCROS_electron2018_bin5
60 prop_binCROS_electron2017_bin1
−2 −1 0 1 2
−10 0 10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



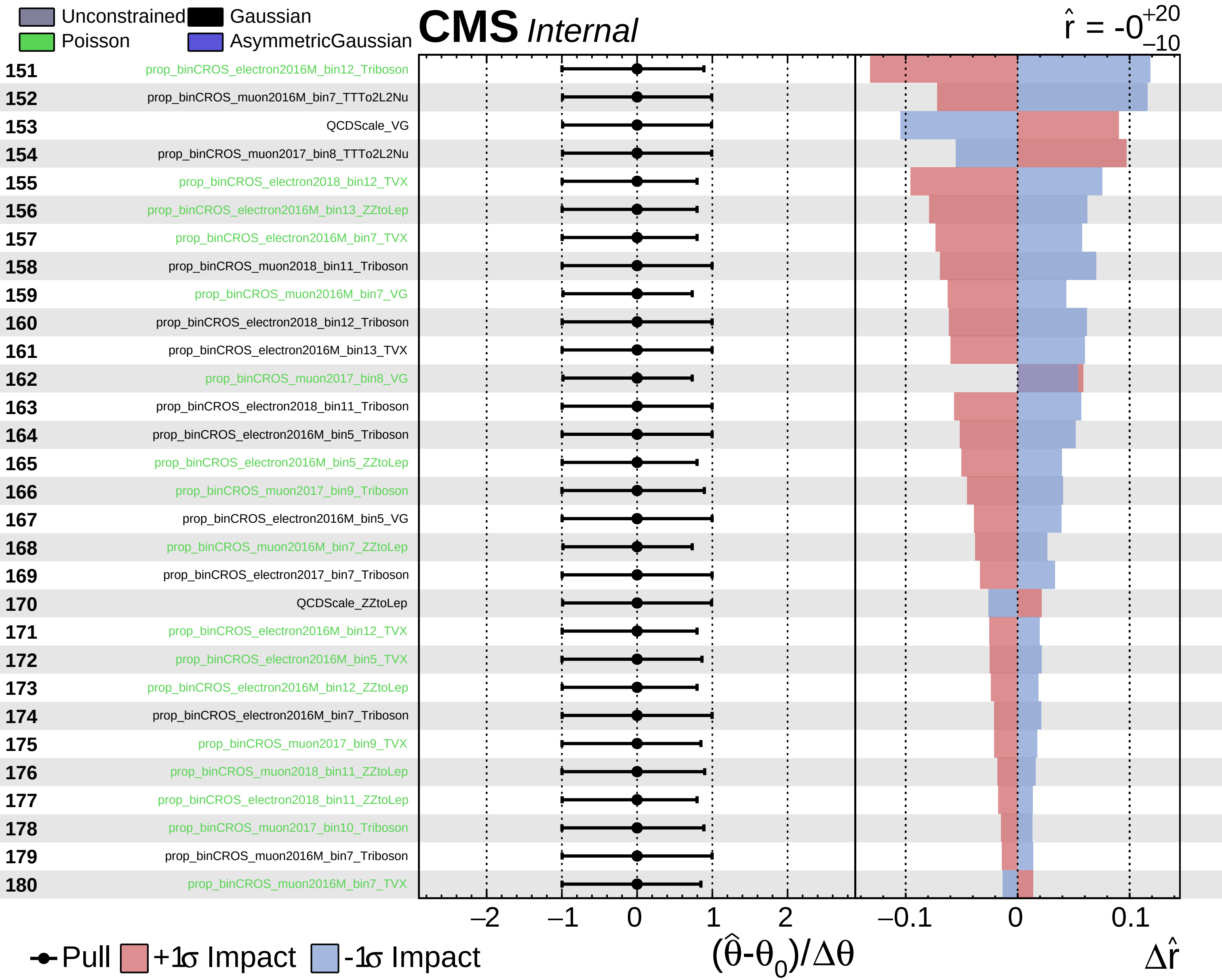
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
61 prop_binCROS_muon2017_bin13
62 prop_binCROS_electron2016M_bin7_WrongSign
63 prop_binCROS_electron2018_bin11_Fake
64 prop_binCROS_muon2016M_bin13
65 prop_binCROS_muon2016M_bin10
66 prop_binCROS_muon2017_bin6
67 prop_binCROS_electron2016M_bin6
68 prop_binCROS_electron2017_bin10
69 prop_binCROS_muon2016M_bin6
70 prop_binCROS_electron2017_bin11
71 prop_binCROS_electron2016M_bin11
72 prop_binCROS_muon2017_bin3
73 jer_2017
74 prop_binCROS_electron2018_bin9
75 OSest_electron_2017
$1.00^{+0.27}_{-0.27}$
76 prop_binCROS_electron2017_bin2
77 mischarge_electron_2017
78 prop_binCROS_electron2017_bin12
79 jer_2016M
80 mischarge_muon_2017
81 mischarge_muon_2016M
82 TES
83 prop_binCROS_muon2017_bin7
84 tau_vsjet_2018
85 prop_binCROS_muon2017_bin8_WrongSign
86 prop_binCROS_muon2017_bin5
87 prop_binCROS_muon2016M_bin11
88 mistag
89 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
90 prop_binCROS_muon2018_bin10
−2
−1
0
1
2
−5
0
5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



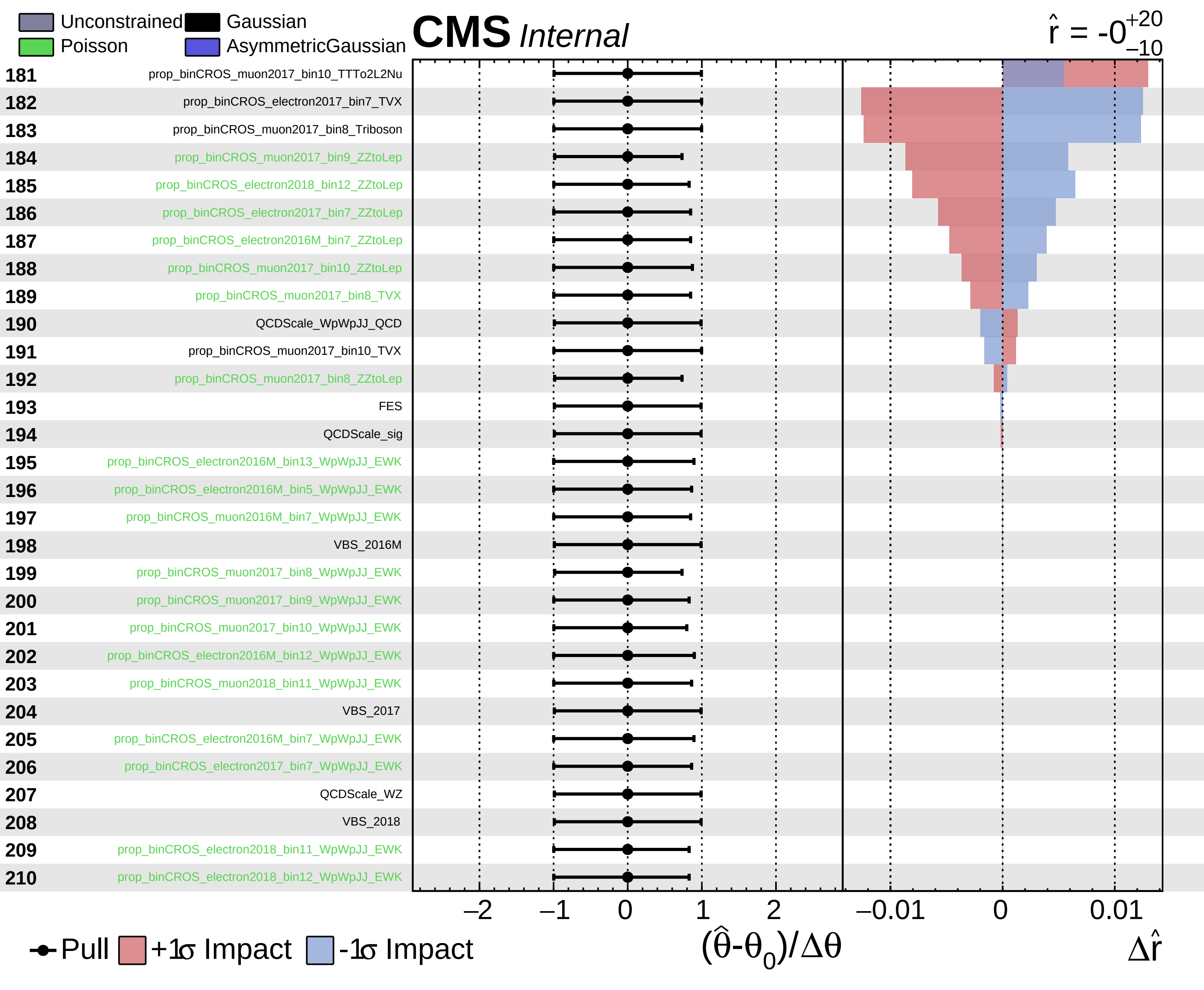
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
91 prop_binCROS_electron2016M_bin10
92 prop_binCROS_electron2017_bin7_Fake
93 prop_binCROS_muon2017_bin1
94 prop_binCROS_muon2018_bin7
95 prop_binCROS_electron2016M_bin5_Fake
96 pdf_Tot
97 metUnclust
98 prop_binCROS_electron2017_bin5
99 prop_binCROS_muon2018_bin6
100 tau_vsjet_2017
101 mischarge_muon_2018
102 prop_binCROS_muon2018_bin11_Fake
103 prop_binCROS_muon2018_bin13
104 OSest_muon_2016M
$1.00^{+0.28}_{-0.28}$
105 jes_2016M
106 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
107 prop_binCROS_muon2018_bin8
108 prop_binCROS_muon2017_bin4
109 prop_binCROS_electron2017_bin6
110 prop_binCROS_electron2018_bin11_TTTo2L2Nu
111 prop_binCROS_muon2016M_bin7_WrongSign
112 prop_binCROS_muon2017_bin2
113 prop_binCROS_electron2017_bin3
114 prop_binCROS_muon2018_bin9
115 prop_binCROS_muon2016M_bin9
116 lumi_2017
117 prop_binCROS_muon2018_bin11_TTTo2L2Nu
118 prop_binCROS_muon2016M_bin3
119 prop_binCROS_muon2016M_bin4
120 prop_binCROS_muon2017_bin9_Fake
−2 −1 0 1 2
−4 −2 0 2 4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



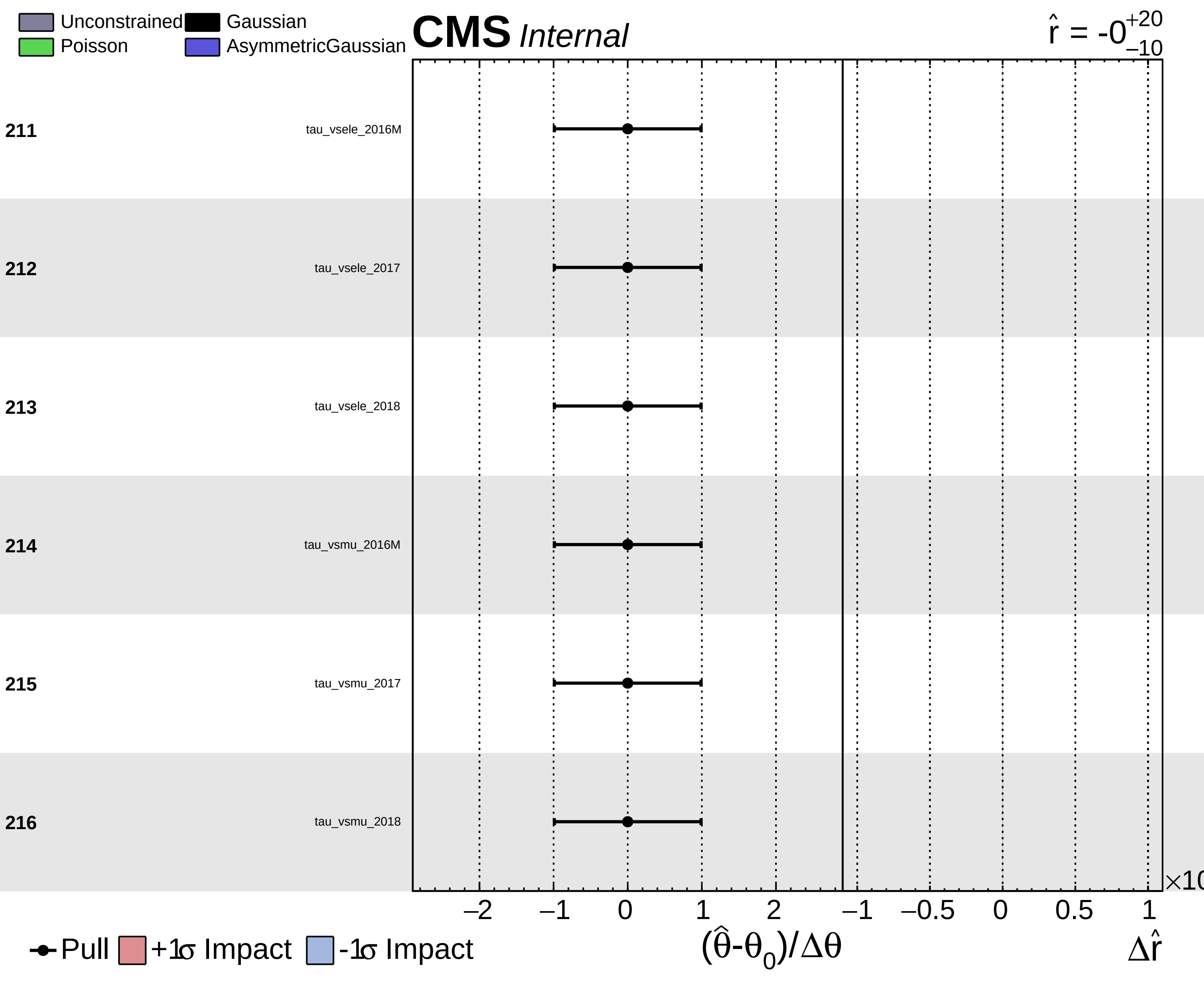
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
121 prop_binCROS_electron2017_bin9
122 prop_binCROS_electron2017_bin4
123 prop_binCROS_muon2017_bin10_WrongSign
124 prop_binCROS_electron2018_bin12_Fake
125 puID
126 lumi_2018
127 prop_binCROS_muon2017_bin11
128 lep_2016M
129 QCDScale_TVX
130 prop_binCROS_muon2016M_bin12
131 prop_binCROS_electron2016M_bin7_Fake
132 prop_binCROS_muon2017_bin8_Fake
133 lumi_2016M
134 prop_binCROS_muon2017_bin9_TTTo2L2Nu
135 prop_binCROS_electron2018_bin12_VG
136 prop_binCROS_electron2017_bin7_TTTo2L2Nu
137 lep_2018
138 prop_binCROS_electron2017_bin8
139 QCDScale_Triboson
140 prop_binCROS_muon2017_bin10_Fake
141 prop_binCROS_electron2016M_bin13_Triboson
142 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
143 prop_binCROS_electron2016M_bin5_WZ
144 prop_binCROS_muon2016M_bin7_Fake
145 prop_binCROS_electron2018_bin12_TTTo2L2Nu
146 prop_binCROS_electron2017_bin7_WZ
147 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
148 prop_binCROS_muon2018_bin11_TVX
149 lep_2017
150 prop_binCROS_electron2018_bin11_TVX
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
151 prop_binCROS_electron2016M_bin12_Triboson
152 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
153 QCDScale_VG
154 prop_binCROS_muon2017_bin8_TTTo2L2Nu
155 prop_binCROS_electron2018_bin12_TVX
156 prop_binCROS_electron2016M_bin13_ZZtoLep
157 prop_binCROS_electron2016M_bin7_TVX
158 prop_binCROS_muon2018_bin11_Triboson
159 prop_binCROS_muon2016M_bin7_VG
160 prop_binCROS_electron2018_bin12_Triboson
161 prop_binCROS_electron2016M_bin13_TVX
162 prop_binCROS_muon2017_bin8_VG
163 prop_binCROS_electron2018_bin11_Triboson
164 prop_binCROS_electron2016M_bin5_Triboson
165 prop_binCROS_electron2016M_bin5_ZZtoLep
166 prop_binCROS_muon2017_bin9_Triboson
167 prop_binCROS_electron2016M_bin5_VG
168 prop_binCROS_muon2016M_bin7_ZZtoLep
169 prop_binCROS_electron2017_bin7_Triboson
170 QCDScale_ZZtoLep
171 prop_binCROS_electron2016M_bin12_TVX
172 prop_binCROS_electron2016M_bin5_TVX
173 prop_binCROS_electron2016M_bin12_ZZtoLep
174 prop_binCROS_electron2016M_bin7_Triboson
175 prop_binCROS_muon2017_bin9_TVX
176 prop_binCROS_muon2018_bin11_ZZtoLep
177 prop_binCROS_electron2018_bin11_ZZtoLep
178 prop_binCROS_muon2017_bin10_Triboson
179 prop_binCROS_muon2016M_bin7_Triboson
180 prop_binCROS_muon2016M_bin7_TVX
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
181 prop_binCROS_muon2017_bin10_TTTo2L2Nu
182 prop_binCROS_electron2017_bin7_TVX
183 prop_binCROS_muon2017_bin8_Triboson
184 prop_binCROS_muon2017_bin9_ZZtoLep
185 prop_binCROS_electron2018_bin12_ZZtoLep
186 prop_binCROS_electron2017_bin7_ZZtoLep
187 prop_binCROS_electron2016M_bin7_ZZtoLep
188 prop_binCROS_muon2017_bin10_ZZtoLep
189 prop_binCROS_muon2017_bin8_TVX
190 QCDScale_WpWpJJ_QCD
191 prop_binCROS_muon2017_bin10_TVX
192 prop_binCROS_muon2017_bin8_ZZtoLep
193 FES
194 QCDScale_sig
195 prop_binCROS_electron2016M_bin13_WpWpJJ_EWK
196 prop_binCROS_electron2016M_bin5_WpWpJJ_EWK
197 prop_binCROS_muon2016M_bin7_WpWpJJ_EWK
198 VBS_2016M
199 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
200 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
201 prop_binCROS_muon2017_bin10_WpWpJJ_EWK
202 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
203 prop_binCROS_muon2018_bin11_WpWpJJ_EWK
204 VBS_2017
205 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
206 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
207 QCDScale_WZ
208 VBS_2018
209 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
210 prop_binCROS_electron2018_bin12_WpWpJJ_EWK
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0^{+20}_{-10}$
211
tau_vsele_2016M
212
tau_vsele_2017
213
tau_vsele_2018
214
tau_vsmu_2016M
215
tau_vsmu_2017
216
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
$\times 10$
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$